M

Thm. Let K be a field, and $K_1, K_2 \leq K$ be subfields.
Define a Composite field of K_1 and K_2 as:

$$K_1 K_2 \overset{\text{def}}{=} \bigcap_{\substack{F \leq K \\ K_1, K_2 \leq K}} F$$

That is, the smallest subfield of K containing $K_1 \cup K_2$.

Lemma. Let K/F be a extension field, and let $\alpha_1, \alpha_2 \in K$. Then,

$$F(\alpha_1) F(\alpha_2) = F(\alpha_1, \alpha_2)$$

Proof. Since $F(\alpha_1, \alpha_2)$ contains both $F(\alpha_1)$ and $F(\alpha_2)$, $\subseteq$, and since $\alpha_1, \alpha_2 \in F(\alpha_1)F(\alpha_2)$, so proved.

Thm. Let K/F be an Extension, and $K_1, K_2 \leq K$ be finite extensions for F. Then,

$$[K_1 K_2 : F] \leq [K_1 : F] \cdot [K_2 : F]$$

Moreover, if $[K_1 : F], [K_2 : F]$ are relatively prime, then the equality hold.
Proof. Let $\{\alpha_1, \cdots, \alpha_n\}$ be a F-basis for K_1, and $\{\beta_1, \cdots, \beta_m\}$ be a F-basis for K_2.
Then, Clearly,

$$K_1 K_2 = F(\alpha_1, \cdots, \alpha_n, \beta_1, \cdots, \beta_m).$$
$$= K_1(\beta_1, \cdots, \beta_m)$$

That is, $K_1 K_2$ is spanned by $\beta_1, \cdots, \beta_m$ over K_1 hence

$$[K_1 K_2 : K_1] \leq m = [K_2 : F].$$

Now, multiplying $[K_1 : F]$ to the both side, we obtain the first assertion.
And, take $[K_1 : F] = n$ and $[K_2 : F] = m$. Since

$$[K_1 K_2 : F] = [K_1 K_2 : K_i][K_i : F] \quad (i = 1, 2)$$

so $[K_1 K_2 : F]$ is divided by n and m, so $\mathrm{lcm}(n,m) \mid [K_1 K_2 : F]$.
And, if $\gcd(n,m) = 1$, then $\mathrm{lcm}(n,m) = 1$, so $nm \leq [K_1 K_2 : F]$.

「Example,

Let $K = \mathbb{Q}(\sqrt[4]{2}, \sqrt{3}, \sqrt{6})$ and $K_1 = \mathbb{Q}(\sqrt[4]{2}, \sqrt{3})$ and $K_2 = \mathbb{Q}(\sqrt{6})$.

Then, $K = K_1K_2$. And, $[K_1 : \mathbb{Q}] = 8$, and $[K_2 : \mathbb{Q}] = 2$.

As we expect, $[K : \mathbb{Q}] = 8 \cdot 2 = 16$?

Since K_1 is $\mathrm{span}_{\mathbb{Q}}\{1, \sqrt[4]{2}, \sqrt{2}, \sqrt[4]{8}, \sqrt{3}, \sqrt[4]{2}\sqrt{3}, \sqrt{6}, \sqrt[4]{8}\sqrt{3}\}$,

and K_2 is $\mathrm{span}_{\mathbb{Q}}\{1, \sqrt{6}\}$,

So $K_2 \subseteq K_1$, K_1K_2 has degree 8.

$$
\begin{tikzcd}
(\leq 2) & K_1K_2 & (\leq 6) \\
K_1 \arrow[ur, dash, "1"] & & K_2 \arrow[ul, dash, "4"'] \\
& \mathbb{Q} \arrow[ul, dash, "8"] \arrow[ur, dash, "2"'] &
\end{tikzcd}
$$